

Physical Activity & BMI Percentile

Research Question

Does physical activity affect BMI percentile among high school students?

Method

- Descriptive Statistics
- Correlation Analysis
- Linear Regression
- Gender Comparison Analysis (Extended)

Variable

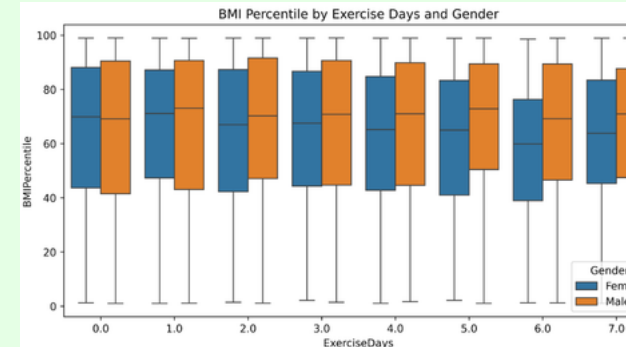
Exercise Days Independent
Gender Independent
BMI Percentile Dependent

Interpretation

- Students who exercised more frequently tended to have slight lower BMI percentile values
- However, the relationship was extremely weak and not statistically significant
- Gender-based analysis showed slight differences in BMI percentile distributions between male and female students
- Gender may influence the relationship between physical activity and BMI percentile

Key Result

n 12,527
r -0.013
Slope -0.14
R² 0.00
P-value 0.145



Regression Equation
 $\text{BMI Percentile} = 65.2935 - 0.1393 \times \text{Exercise Days}$

Very weak negative relationship

Exercise frequent explains almost none of the variation in BMI percentile

Conclusion

- No significant relationship was found between physical activity and BMI percentile
- Gender differences were observed in BMI percentile distributions
- Gender may influence the relationship between physical activity and BMI percentile

Exercise & BMI

